

A subsampling MCMC approach for Bayesian model selection and model averaging

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Subsampling MCMC for Bayesian Variable Selection and Model Averaging

- ▶ Background
 - ▶ Bayesian Generalized linear Models (BGLM)
 - ▶ Mode Jumping MCMC (MJMCMC)
 - ▶ Bayesian Generalized Nonlinear Models (BGNLM)
 - ▶ Genetically Modified (Mode Jumping) MCMC (GMJMCMC)
 - ▶ The marginal posterior model probabilities
 - ▶ Posterior marginal probability for covariates and features
 - ▶ Bayesian model selection and averaging

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- ▶ Contribution
 - ▶ Subsampling in estimating marginal likelihood
 - ▶ (MJ)MCMC becomes time inhomogeneous MCMC (TIMCMC) with theoretical guarantees
 - ▶ TIMCMC within GMJMCMC

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 - ▶ Subsampling in estimating marginal likelihood
 - ▶ (MJ)MCMC becomes time inhomogenous MCMC (TIMCMC) with theoretical guarantees
 - ▶ TIMCMC within GMJMCMC
- ▶ Demonstration
 - ▶ Variable selection in magic telescope data.
 - ▶ Simulation studies in linear regressions and **tall data**.
 - ▶ Simulation study with logistic logic regression and **tall data**.
 - ▶ Abalone shell predictions.
 - ▶ Heart issues prediction with **tall data**.

Bayesian Generalized linear Model

The **model** is specified:

$$Y_i | \mu_i, \phi \sim f(y | \mu_i, \phi), i = 1, \dots, n; \quad (1)$$

$$h(\mu_i) = \beta_0 + \sum_{j=1}^p \gamma_j \beta_j x_j \quad (2)$$

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- ▶ $f(\cdot | \mu, \phi)$ - density from exponential family with mean μ_i and dispersion parameter ϕ ;
- ▶ h - link function;
- ▶ **covariates:** $x_j, j = 1, \dots, p$
- ▶ $\beta_j \in \mathbb{R}$ - regression coefficients of j -th covariate;
- ▶ $\gamma_j \in \{0, 1\}$ - indicator variable for j -th covariate.
- ▶ For this talk we are interested also in **large n !**
- ▶ Priors discussed later.

Bayesian Generalized Nonlinear Model (Hubin, Storvik, and Frommlet 2021a)

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- ▶ **features:** $F_j(\mathbf{x}_i, \alpha_j)$, $j = 1, \dots, q$
- ▶ $\beta_j \in \mathbb{R}$ - regression coefficients of j -th feature;
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Hierarchy of the features

A feature $F_j(\mathbf{x}, \alpha_j), j \in \{p+1, \dots, q\}$ can be constructed recursively through:

$$F_j(\mathbf{x}, \alpha_j) = v_j(\alpha_j^T \vec{F}_{1:j-1}(\mathbf{x}))$$

$v_j \in \mathcal{G}$ is one of the allowed basic function from set $\mathcal{G}$.

Types and meaning of functions in $\mathcal{G}$

- ▶ Neural Networks: $\text{logit}(x)$, $\tanh(x)$, $\text{erf}(x)$, $\text{ReLU}(x)$;
- ▶ Polynomials: $F_k(\mathbf{x}) * F_l(\mathbf{x}) = \exp(\log(F_k(\mathbf{x})) + \log(F_l(\mathbf{x})))$;
- ▶ CART: $I(x \geq 1)$;
- ▶ MARS: $\max\{0, x - t\}$ and $\max\{0, t - x\}$;
- ▶ Fractional polynomials: $x^{\frac{1}{a}} = \exp(b \log(x))$, $b = \frac{1}{a}$;
- ▶ **Logical AND, OR and NOT:** $L_k \wedge L_l = L_k * L_l$,
 $L_k \vee L_l = L_k + L_l - L_k * L_l$, **and** $\bar{L}_k = 1 - L_k$.
- ▶ Symbolic regressions.

Features are **functional** trees (Hubin, Storvik, and Frommlet 2021a)

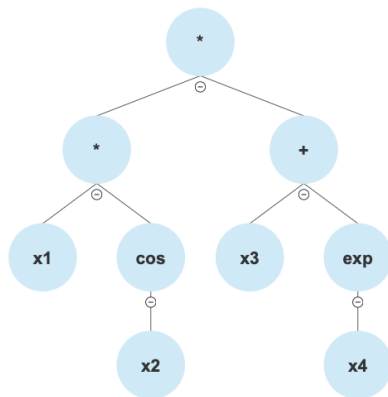


Figure: Functional tree termed *feature* $F = (x_1 * \cos(x_2)) * (x_3 + \exp(x_4))$

Depth of trees constrained to make q finite!

Prior specification

Model priors

Model topology defined by $\mathfrak{m} = (\gamma_1, \dots, \gamma_q)$

Priors of $\mathfrak{m}$ should guarantee regularization (parsimonious models)

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$$p(\mathbf{m}) \propto \mathbb{I}(|\gamma_{1:q}| \leq Q) \prod_{j=1}^q a^{\gamma_j c(F_j(\mathbf{x}, \alpha_j))}, \quad 0 < a < 1 \quad (5)$$

- ▶ $c(F_j(\mathbf{x}, \alpha_j)) \geq 0$ non-decreasing **complexity measure** of feature
- ▶ Q is the maximal number of features per given model.

Example: 1 + number of algebraic operations in a feature;

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Parameter priors

Priors for β_j (and ϕ): Problem-specific and computational considerations

Conjugate priors, Jeffreys priors, robust g priors, etc.

Computation of model posterior

Assume first that marginal likelihood

$$p(\mathcal{D}|\mathbf{m}) = \int_{\theta_{\mathbf{m}} \in \Theta_{\mathbf{m}}} p(\mathcal{D}|\theta_{\mathbf{m}}, \mathbf{m})p(\theta_{\mathbf{m}}|\mathbf{m})d\theta_{\mathbf{m}} \text{ known for all models}$$

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Denominator

Model space $\mathcal{M}$ extremely **large and multimodal**

- Markov Chain Monte Carlo estimator (**MC**) at iteration t

$$\hat{p}_{MC}^{(t)}(\mathbf{m}|\mathcal{D}) = t^{-1} \sum_{s=1}^t \mathbb{I}(\mathbf{m}^{(s)} = \mathbf{m}). \quad (7)$$

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- ▶ or renormalized estimator (**RM**) at iteration t

$$\hat{p}_{RM}^{(t)}(\mathbf{m}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{m})p(\mathbf{m})}{\sum_{\mathbf{m}' \in \mathcal{M}_V^{(t)}} p(\mathcal{D}|\mathbf{m}')p(\mathbf{m}')} \mathbb{I}(\mathbf{m} \in \mathcal{M}_V^{(t)}). \quad (8)$$

Posterior marginal probability (PMP) for features

Marginal posterior of models to posterior marginal probability for features

- ▶ Monte Carlo estimator

$$\hat{p}_{MC}^{(t)}(\gamma_j = 1|\mathcal{D}) = \sum_{\mathbf{m} \in \mathcal{M}_V^{(t)}} \mathbb{I}(\gamma_j = 1) \hat{p}_{MC}^{(t)}(\mathbf{m}|\mathcal{D})$$

- ▶ Renormalized estimator

$$\hat{p}_{RM}^{(t)}(\gamma_j = 1|\mathcal{D}) = \sum_{\mathbf{m} \in \mathcal{M}_V^{(t)}} \mathbb{I}(\gamma_j = 1) \hat{p}_{RM}^{(t)}(\mathbf{m}|\mathcal{D})$$

- ▶ Similarly, BMA for predictions through RM and MC estimators.

Algorithmic Details. **MJMCMC**

Multiple modes result in poor mixing of standard MCMC

Mode jumping MCMC (MJMCMC) (Hubin and Storvik 2018a; Tjelmeland and Hegstad 2001)

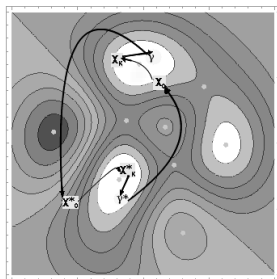
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- ▶ After certain number of MCMC steps make a jump
Random model plus local improvement creates new proposal;
- ▶ Get valid acceptance probability for Metropolis-Hastings
(Theorem 1 in Hubin and Storvik 2018a).



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- ▶ MJMCMC is embedded in a genetic algorithm which updates a finite population of predictors of size $Q \leq d \ll q$;
- ▶ Populations $\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_{T_{max}}$ each $\mathcal{S}_t$ is a set of d predictors;
- ▶ Use evolutionary dynamic to generate further populations.
 $\mathcal{M}^* \rightarrow \mathcal{M}$ (Theorem 1 in Hubin, Storvik, and Frommlet 2020 and Theorem 1 in Hubin, Storvik, and Frommlet 2021a)
- ▶ Parallelize in as many chains as one want, then combine unique models in the RM estimators.

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RGMJMCMC Solution Hubin, Storvik, and Frommlet 2021b

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The marginal posterior

- ▶ (MJ)MCMC **requires** the marginal likelihoods $p(\mathcal{D}|\mathbf{m})$
- ▶ Expensive to estimate MLIK through simulation
- ▶ **Laplace approximation** is way faster, but **still slow for large n** as it requires the posterior mode of the model parameters:

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Let $\hat{\theta}_{\mathbf{m}}$ denote the posterior mode. Then, the approximation is :

$$\hat{p}_{\hat{\theta}_{\mathbf{m}}}(\mathcal{D}|\mathbf{m}) = p(\mathcal{D}|\hat{\theta}_{\mathbf{m}}, \mathbf{m})p(\hat{\theta}_{\mathbf{m}}|\mathbf{m})(2\pi)^{|\theta_{\mathbf{m}}|/2} \left| S(\hat{\theta}_{\mathbf{m}}, \mathbf{m}) \right|^{-1/2}, \quad (9)$$

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where $S(\hat{\theta}_{\mathbf{m}}, \mathbf{m})$ is the Hessian.

- ▶ The integration (9) becomes exact for a model with Gaussian observations and a normal prior for the coefficients β ;
- ▶ Otherwise, it would have an error of order of at least $\mathcal{O}(n^{-1})$, which **diminishes as n increases (good news)**;
- ▶ **But for large n computing $\hat{\theta}_{\mathbf{m}}$ is hard using full sample optimization (bad news).**

Contribution

- ▶ Subsampling optimization in getting $\tilde{\theta}_m^{(t)} \rightarrow \hat{\theta}_m$ as $t \rightarrow \infty$
- ▶ Reestimation for new visits of the same model
- ▶ Induces a time inhomogenous MCMC (TIMCMC) for MJMCMC since the transition kernel changes with t
- ▶ Allows us to also have TIMCMC within GMJMCMC

Subsampling

Without subsampling

- ▶ The mode $\hat{\theta}_m$ is obtained through optimization (Newton, Gradient Descent)
- ▶ Each iteration goes through all data

With subsampling

- ▶ The **approximate** posterior mode $\tilde{\theta}_m^{(t)}$ can be obtained quicker through subsampling (SGD)
- ▶ Using only a subsample of the data at each iteration reduces complexity
- ▶ A final step calculating the likelihood on all data gives $0 < \hat{p}_{\tilde{\theta}_m^{(t)}}(\mathcal{D}|\mathbf{m}) \leq \hat{p}_{\hat{\theta}_m}(\mathcal{D}|\mathbf{m}), \forall t$, but $\hat{p}_{\tilde{\theta}_m^{(t)}}(\mathcal{D}|\mathbf{m}) \rightarrow \hat{p}_{\hat{\theta}_m}(\mathcal{D}|\mathbf{m})$ as, $t \rightarrow \infty$.

Time Inhomogenous MCMC

Algorithm 1: The TIMCMC algorithm

Select a model $\mathbf{m}^{(0)}$, put $\tilde{\boldsymbol{\theta}}_{\mathbf{m}}^{(0)} = \text{init-mode}(\mathbf{m}^{(0)})$ and $\mathcal{M}_V = \{\mathbf{m}^{(0)}\}$;

for $t = 1, 2, \dots$ **do**

 Select a model $\mathbf{m}^*$ from some proposal distribution $q(\mathbf{m}^*|\mathbf{m}^{(t-1)})$;

if $\mathbf{m}^* \notin \mathcal{M}_V$ **then**

 Put $\tilde{\boldsymbol{\theta}}_{\mathbf{m}^*}^{(t)} \leftarrow \text{init-mode}(\mathbf{m}^*)$ and $\mathcal{M}_V \leftarrow \{\mathcal{M}_V, \mathbf{m}^*\}$;

else

 Put $\tilde{\boldsymbol{\theta}}_{\mathbf{m}^*}^{(t)} \leftarrow \text{update-mode}(\mathbf{m}^*, \tilde{\boldsymbol{\theta}}_{\mathbf{m}^*}^{(t-1)})$;

end

 Set the next state of the chain to

$$\mathbf{m}^t = \begin{cases} \mathbf{m}^* & \text{with probability } \alpha = r_{mh}^*(\mathbf{m}^{t-1}, \mathbf{m}^*); \\ \mathbf{m}^{t-1} & \text{otherwise;} \end{cases}$$

where

$$r_{mh}^*(\mathbf{m}, \mathbf{m}^*) = \min \left\{ 1, \frac{p(\mathbf{m}^*)\hat{p}_{\tilde{\boldsymbol{\theta}}_{\mathbf{m}^*}^{(t)}}(\mathcal{D}|\mathbf{m}^*)q(\mathbf{m}|\mathbf{m}^*)}{p(\mathbf{m})\hat{p}_{\tilde{\boldsymbol{\theta}}_{\mathbf{m}}^{(t)}}(\mathcal{D}|\mathbf{m})q(\mathbf{m}^*|\mathbf{m})} \right\},$$

where $q(\cdot|\cdot)$ is the proposal of the addressed MCMC.

end

Remark

- ▶ Works with *any optimisation* method which *converges*: $\tilde{\boldsymbol{\theta}}_{\mathbf{m}}^{(t)} \rightarrow \hat{\boldsymbol{\theta}}_{\mathbf{m}}$ as $t \rightarrow \infty$ in, say, probability or almost surely.
- ▶ The use of auxiliary variables for the generation of $\mathbf{m}^*$ is often beneficial: we use MJMCMC (Hubin and Storvik 2018b).

TIMCMC for (G)MJMCMC

Theorem 1 (Lachmann et al. 2022)

- ▶ Assume that the estimates generate a sequence $\{\tilde{\theta}_m^{(t)}\}$ such that $\hat{p}_{\tilde{\theta}_m^{(t)}}(\mathcal{D}|\mathbf{m}) \geq \hat{p}_{\tilde{\theta}_m^{(t-1)}}(\mathcal{D}|\mathbf{m}) > 0$ and the sequence converges almost surely **(a)** or in probability **(b)**.
 - ▶ Then, the Markov chain is irreducible and recurrent.
 - ▶ For the renormalized estimator it holds that
- (a) $\hat{p}_{\tilde{\theta}_m^{(t)},RM}^{(t)}(\mathbf{m}|\mathcal{D}) \xrightarrow{a.s.} p_{\hat{\theta}_m, RM}^{(t)}(\mathbf{m}|\mathcal{D})$ as $t \rightarrow \infty$
- (b) $\hat{p}_{\tilde{\theta}_m^{(t)},RM}^{(t)}(\mathbf{m}|\mathcal{D}) \xrightarrow{P} p_{\hat{\theta}_m, RM}^{(t)}(\mathbf{m}|\mathcal{D})$ as $t \rightarrow \infty$.

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And for stricter conditions (require additionally A3 from Fort, Moulines, and Priouret 2011) we have convergence of *MC* estimates outlined in **Theorem 2 in Lachmann et al. 2022**.

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- ▶ We can use TIMCMC instead of MJMCMC in GMJMCMC (Lachmann and Hubin 2023)
- ▶ Our R package FBMS supports this

Demonstration

Demonstration optimisation algorithm: S-IRLS-SGD $\rightarrow$ TIMCMC

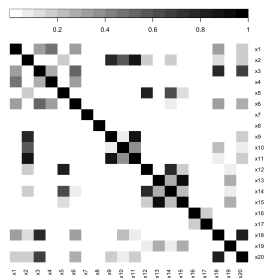
Remark

Any stochastic optimisation can be used as $\text{opt}()$ here, but we recommend a version of batch stochastic gradient decent (SGD) with subsample iterative reweighed least squares (S-IRLS) used for initialisation resulting in a S-IRLS-SGD procedure, see Lachmann et al. 2022 for details.

- ▶ Simulation study on linear models with tall data
- ▶ Study of magic telescope data;
- ▶ Simulation study on logistic logic regression with tall data.
- ▶ Abalone shell predictions.
- ▶ Heart issues classification with tall data.

Simulation (linear model)

- Variables $\mathbf{x} \in \mathcal{R}^{15}$



- $\beta = (0.48, 8.72, 1.76, 1.87, 0, 0, 0, 0, 4, 0, 0, 0, 0, 0, 0)/\sqrt{n/100}$.
- Dependent variable $\mathbf{y}$ for $n = 100,000$ observations generated from:

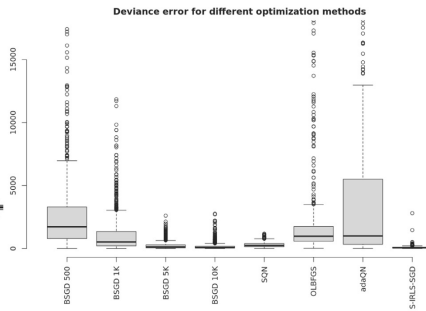
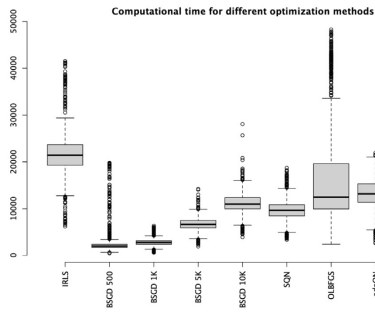
$$y_i \sim N(\mathbf{x}_i^T \beta, 1). \quad (10)$$

Different optimizers, 128 best models

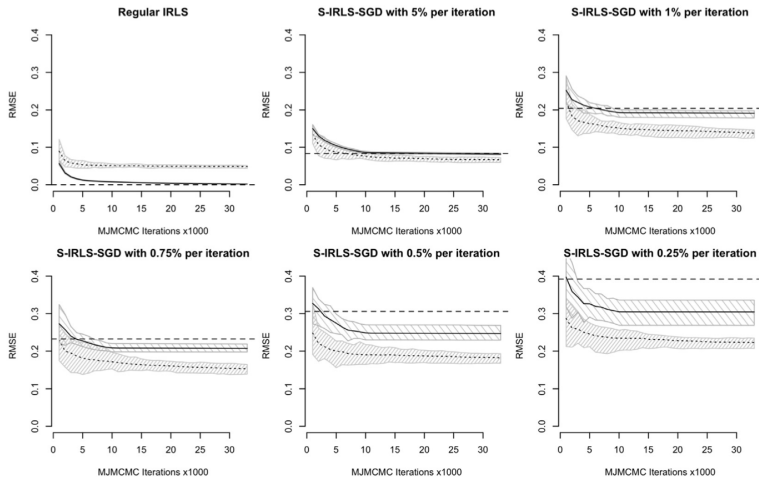
Using g priors

$$\log p(\mathbf{y}|\mathbf{m}) = \text{const} + \frac{n-p}{2} \log(1+g) - \frac{n-1}{2} \log(1+g(1-R_m^2))$$

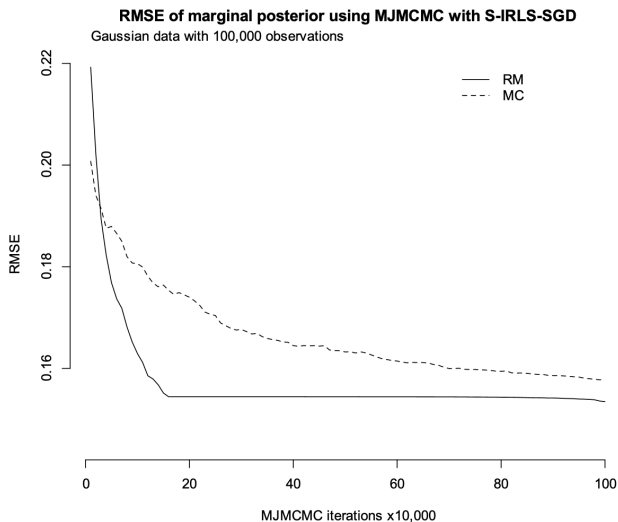
We still need posterior modes



Simulation (linear model, many subsample sizes), g-prior



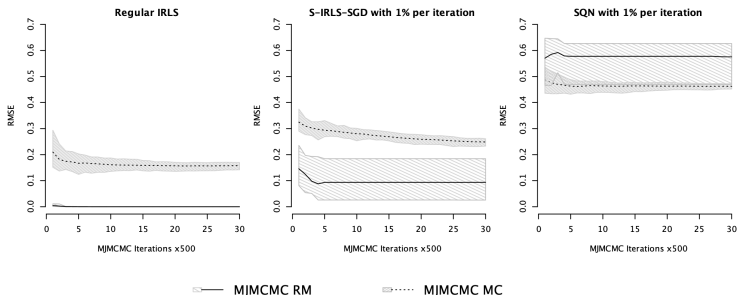
Simulation (linear model, subsample size of 0.75%), g-prior



Performance under different optimisers

MAGIC Gamma Telescope Data from UCI, $n = 19020$, $p = 20$, Jeffreys prior.

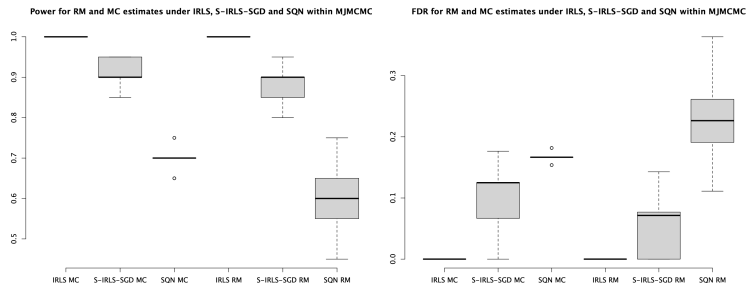
RMSE of marginal posterior using MJMCMC, real logistic data



RMSE compared to the true distribution for MJMCMC using both regular IRLS, S-IRLS-SGD and SQN with a subsample size of 1% per iteration. MJMCMC was run 20 times and for 15 000 iterations for every algorithm. Stats based on 20 runs.

Performance under different optimisers

MAGIC Gamma Telescope Data from UCI <https://archive.ics.uci.edu/ml/datasets/magic+gamma+telescope>.



Power and FDR for MJMCMC combined with IRLS, S-IRLS-SGD, and stochastic quasi-Newton (SQN) optimisation as `opt()` in `update-mode()`.

Simulation (logic regression, tall data), Jeffreys prior

- ▶ 50 binary covariates, $n = 100,000$, $\mathbf{x} \stackrel{iid}{\sim} \text{Bernoulli}(0.5)$
- ▶ Dependent variable $\mathbf{y} \sim \text{N}(\mathbf{x}^T \boldsymbol{\beta}, 1)$
- ▶ $\boldsymbol{\beta} = (1, 1.5, 1.5, 6.6, 3.5, 9, 7, 7, 7)$ for the data generative features and 0 otherwise

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- ▶ $\mathbf{y}$ was specified as
$$E(\mathbf{y}|\mathbf{x}, \boldsymbol{\beta}) = \beta_0 + \beta_1 \mathbf{x}_7 + \beta_2 \mathbf{x}_8 + \beta_3 \mathbf{x}_{18} \mathbf{x}_{21} + \beta_4 \mathbf{x}_2 \mathbf{x}_9 + \beta_5 \mathbf{x}_{12} \mathbf{x}_{20} \mathbf{x}_{37} + \beta_6 \mathbf{x}_1 \mathbf{x}_3 \mathbf{x}_{27} + \beta_7 \mathbf{x}_4 \mathbf{x}_{10} \mathbf{x}_{17} \mathbf{x}_{30} + \beta_8 \mathbf{x}_{11} \mathbf{x}_{13} \mathbf{x}_{19} \mathbf{x}_{50}.$$
- ▶ 0.75% subsample size

Power and false discovery rate
(without subsampling in parentheses)

Threads	Power	FDR
1	0.42 (0.40)	0.89 (0.88)
32	0.67 (0.78)	0.78 (0.50)
128	0.76 (0.93)	0.74 (0.20)

Abalone shell predictions, Jeffreys prior

- ▶ The Abalone dataset, available at <https://archive.ics.uci.edu/ml/datasets/Abalone>
- ▶ $n = 4177$, $n_t = 1000$ kept for testing
- ▶ 10% subsample size in S-IRLS-SGD
- ▶ 32 Threads run in parallel
- ▶ Comparison with and without subsampling
- ▶ Performance is very similar to full sample

Model	RMSE	MAE	CORR
BGNLM_SUB	1.9506 (1.9343, 2.0121)	1.4443 (1.4239, 1.4653)	0.7854 (0.7738, 0.7892)
BGNLM	1.9573 (1.9334, 1.9903)	1.4467 (1.4221, 1.4750)	0.7831 (0.7740, 0.7895)
RFOREST	2.0352 (2.0020, 2.0757)	1.4924 (1.4650, 1.5259)	0.7633 (0.7530, 0.7712)
BGLM	2.0758 (-,-)	1.5381 (-,-)	0.7522 (-,-)
LASSO	2.0765 (-,-)	1.5386 (-,-)	0.7514 (-,-)
VARBAYES	2.0779 (-,-)	1.5401 (-,-)	0.7516 (-,-)
GR	2.0801 (-,-)	1.5401 (-,-)	0.7500 (-,-)
RIDGE	2.1340 (-,-)	1.5649 (-,-)	0.7347 (-,-)

Heart Disease Health Indicators Dataset, Jeffreys prior

- ▶ The dataset, available at <https://www.kaggle.com/datasets/alexteboul/heart-disease-health-indicators-dataset>
- ▶ $n = 253680$, 10 fold CV.
- ▶ 1% subsample size in S-IRLS-SGD
- ▶ 1 thread
- ▶ Comparison with and without subsampling
- ▶ Similar performance to full sample, but time reduced

Model	AUC	ACC	FNR	FPR	TIME
BGNLM	0.8472 (0.8439,0.8536)	0.9076 (0.9060,0.9102)	0.0845 (0.0822,0.0865)	0.4509 (0.4207,0.4782)	4095
LR	0.8464 (0.8433,0.8536)	0.9075 (0.9051,0.9100)	0.0841 (0.0817,0.0859)	0.4555 (0.4315,0.4991)	1
BGNLM_SUB	0.8462 (0.8433,0.8540)	0.9069 (0.9057,0.9104)	0.0854 (0.0825,0.0874)	0.4523 (0.4273,0.4815)	1423
RF	0.8433 (0.8370,0.8496)	0.9075 (0.9061,0.9102)	0.0906 (0.0880,0.0923)	0.3500 (0.3129,0.4053)	329
LASSO	0.8410 (0.8360,0.8461)	0.9067 (0.9057,0.9098)	0.0924 (0.0895,0.0938)	0.2967 (0.2444,0.3608)	2
RIDGE	0.8409 (0.8358,0.8460)	0.9066 (0.9054,0.9096)	0.0926 (0.0898,0.0941)	0.3043 (0.2368,0.3556)	2

Conclusions

- ▶ Introduced subsampling in estimating the marginal likelihoods;
- ▶ Combined with MCMC;
- ▶ Resulted in TIMCMC that converges;
- ▶ Use this TIMCMC for model selection and averaging for BGLM;
- ▶ Extend to GMJCMC for BGNLM;
- ▶ Some good results, but no free lunch in general.

Thank you!

► Questions?



References



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